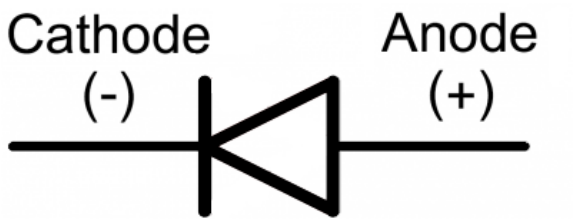
Sekure Networks Best Practices for PCB Design V1.5 070124

1. **Component Placement:**
    - **Pin 1 Marking:** Clearly mark pin 1 on all ICs with a distinctive silkscreen dot or notch.
    - **Designator Placement:** Place the designator (e.g., U1, R1) close to pin 1 for easy identification and orientation during assembly.
    - **Orientation Consistency:** Ensure all similar components are oriented in the same direction to reduce errors during manual assembly.

2. **Silkscreen Markings:**
    - **Readable Text:** Use legible font sizes and avoid placing text under components or on vias and pads.
    - **Polarity Indicators:** Clearly mark the polarity on polarized components like diodes and capacitors.
    - **Reference Designators:** Ensure reference designators are visible after assembly, especially for critical components.
    - **Consistent Designators:** Use consistent designators throughout the entire design to avoid confusion and errors during assembly and testing.

3. **LEDs:**
    - **Cathode Marking:** Clearly mark the cathode on LEDs with a silkscreen line or other indicator.
    - **Polarity Indicators:** Use clear and consistent symbols to indicate the anode and cathode sides of LEDs.
    - **Icon Image:** Use the LED icon image on the silkscreen when space permits to provide clear visual identification of LED components.



4. **Clearances and Spacing:**
    - **Adequate Spacing:** Maintain sufficient spacing between components to prevent shorts and ensure ease of soldering.
    - **Trace Clearance:** Follow the required trace clearances based on the voltage and current ratings.

5. **Thermal Management:**
    - **Thermal Relief Pads:** Use thermal relief pads for through-hole components to facilitate easier soldering and rework.
    - **Heat Dissipation:** Place high-power components where they can dissipate heat effectively, and consider adding heat sinks or thermal vias if necessary.

6. **Routing and Layout:**
    - **Shortest Path:** Route critical signals (e.g., high-speed or sensitive analog signals) using the shortest and most direct paths.
    - **Ground Planes:** Use continuous ground planes to minimize noise and provide a stable reference for signals.
    - **Differential Pairs:** Route differential pairs together and maintain consistent spacing to ensure signal integrity.

7. **Power and Ground:**
    - **Decoupling Capacitors:** Place decoupling capacitors close to the power pins of ICs to filter out noise.
    - **Power Distribution:** Use wide traces or planes for power distribution to reduce voltage drops and improve current handling.

8. **Design for Manufacturability (DFM):**
    - **Test Points:** Include test points for critical signals to facilitate testing and debugging.
    - **Panelization:** Consider panelizing the PCB for efficient manufacturing, especially for smaller boards. Any PCB smaller than 50 mm x 50 mm should be placed on a panel, and mouse bites should be used to allow the PCB to separate from the panel easily.
    - **Component Accessibility:** Ensure components are accessible for probing, rework, and replacement if needed.

9. **Stencil Layer:**
    - **Aperture Reduction:** Create a stencil layer with a 10% reduced aperture to ensure proper solder paste application and prevent bridging or excessive solder.

10. **Documentation and Review:**
    - **Comprehensive Documentation:** Provide detailed assembly drawings, BOM, Pick & Place, DXF, 3D Rendering PDF, and schematic diagrams.
    - **Design Reviews:** Conduct thorough design reviews and DFM checks to catch potential issues before production.

11. **Common Designators:**
    - **Capacitors:** C1, C2, C3, ...
    - **Resistors:** R1, R2, R3, ...
    - **Inductors:** L1, L2, L3, ...
    - **Diodes:** D1, D2, D3, ...
    - **Transistors:** Q1, Q2, Q3, ...
    - **Integrated Circuits:** U1, U2, U3, ...
    - **Connectors:** J1, J2, J3, ...
    - **Test Points:** TP1, TP2, TP3, ...
    - **Switches:** SW1, SW2, SW3, ...
    - **LEDs:** LED1, LED2, LED3, ...